

# Loan Approval Model Evaluation Report

This report presents the performance evaluation and comparison of different Machine Learning models used for Loan Approval Prediction.

## Best Performing Model

| Model     | XGBoost |
|-----------|---------|
| Accuracy  | 0.9377  |
| Precision | 0.8896  |
| Recall    | 0.8215  |
| F1 Score  | 0.8542  |

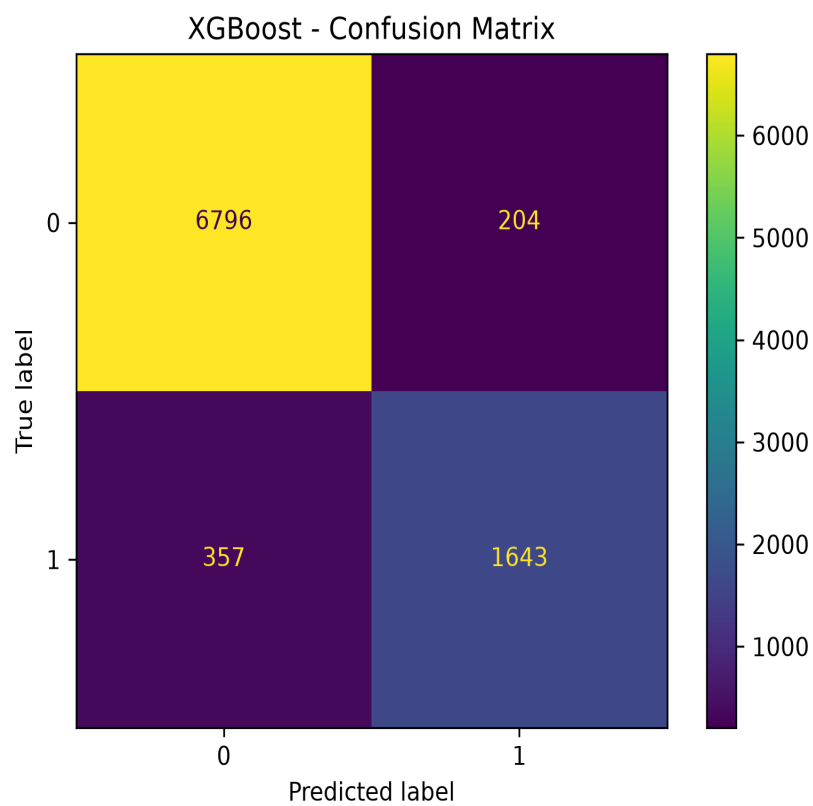
## Model Comparison

| Model               | Accuracy | Precision | Recall | F1 Score |
|---------------------|----------|-----------|--------|----------|
| XGBoost             | 0.9377   | 0.8896    | 0.8215 | 0.8542   |
| Random Forest       | 0.9274   | 0.8918    | 0.7665 | 0.8244   |
| Decision Tree       | 0.8999   | 0.7738    | 0.7765 | 0.7751   |
| Logistic Regression | 0.8993   | 0.7888    | 0.7470 | 0.7673   |

# XGBoost

| Metric    | Score  |
|-----------|--------|
| Accuracy  | 0.9377 |
| Precision | 0.8896 |
| Recall    | 0.8215 |
| F1 Score  | 0.8542 |

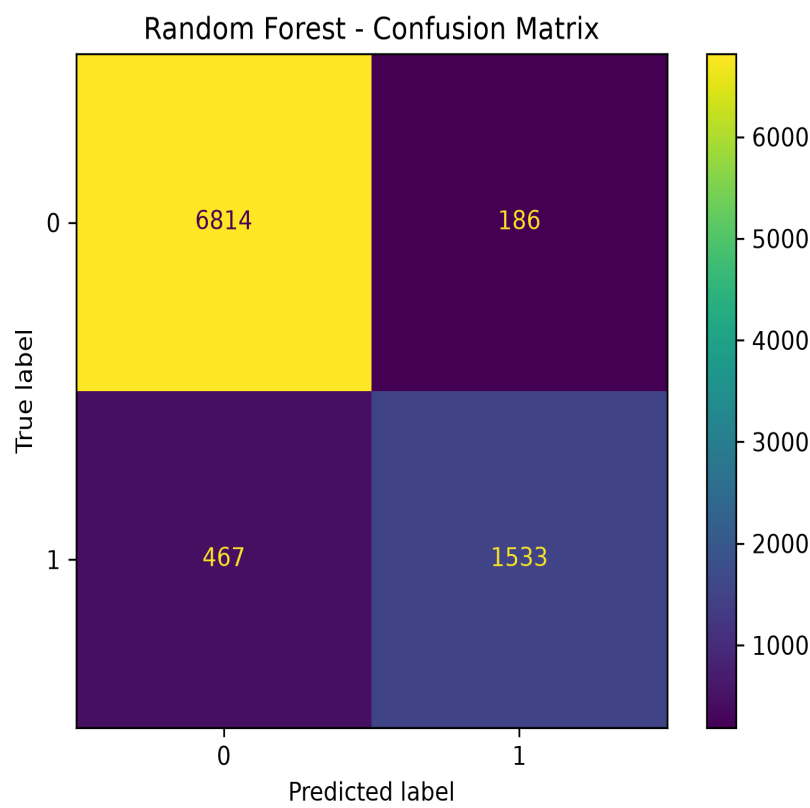
## Confusion Matrix



# Random Forest

| Metric    | Score  |
|-----------|--------|
| Accuracy  | 0.9274 |
| Precision | 0.8918 |
| Recall    | 0.7665 |
| F1 Score  | 0.8244 |

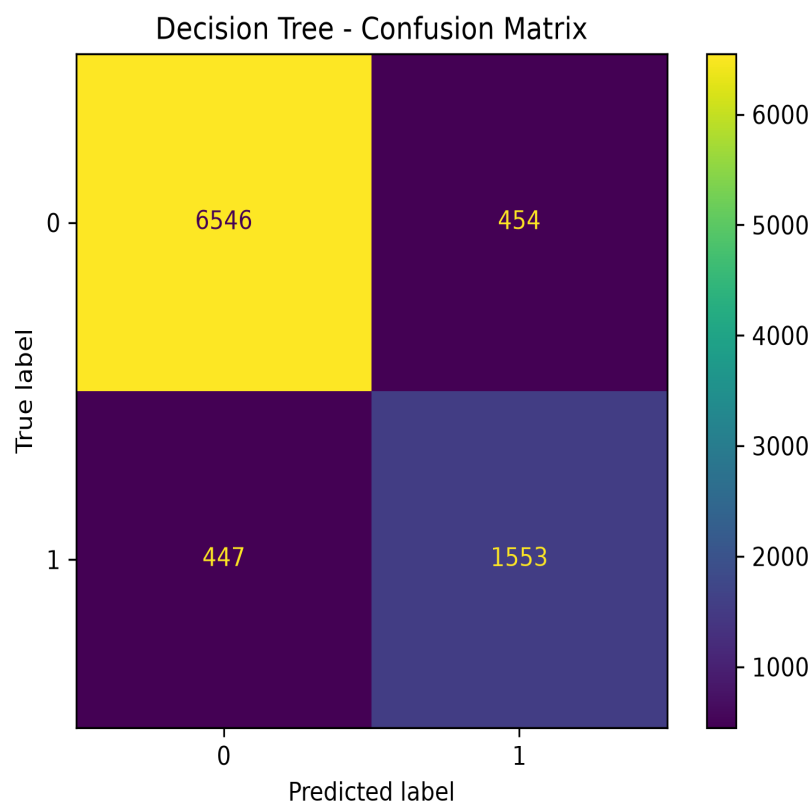
## Confusion Matrix



# Decision Tree

| Metric    | Score  |
|-----------|--------|
| Accuracy  | 0.8999 |
| Precision | 0.7738 |
| Recall    | 0.7765 |
| F1 Score  | 0.7751 |

## Confusion Matrix



# Logistic Regression

| Metric    | Score  |
|-----------|--------|
| Accuracy  | 0.8993 |
| Precision | 0.7888 |
| Recall    | 0.7470 |
| F1 Score  | 0.7673 |

## Confusion Matrix

